

Chapter 2

Linear Equations

Opening Challenge

Solve

$$2026x + 2025 = 2025x + 2026.$$

You could move terms carefully from one side to the other. But before doing that, look at the structure. The coefficients differ by only 1, and so do the constant terms. Subtracting $2025x + 2025$ from both sides gives

$$x = 1.$$

This chapter is about more than solving one equation at a time. We shall ask several questions:

1. Why do the usual steps for solving equations work?
2. Can we derive a general formula for a linear equation?
3. When does an equation have one solution, no solution, or infinitely many solutions?
4. When an equation looks nonlinear, can it still reduce to a linear one?
5. When must a proposed solution be checked in the original equation?

The main theme is simple: an equation is a balance. Whatever operation we perform, we must preserve that balance.

2.1 What Is a Linear Equation?

An equation states that two expressions have the same value. For example,

$$3x - 5 = 10$$

asks for all values of x that make the two sides equal.

A **linear equation in one variable** can be written in the form

$$ax + b = cx + d,$$

where a, b, c, d are constants.

The variable x appears only to the first power. There are no terms such as x^2 , $1/x$, or $\sqrt{x}$ in a linear equation.

The Balance Principle

Suppose

$$A = B.$$

Then the equality remains true if we

- add the same number to both sides,
- subtract the same number from both sides,
- multiply both sides by the same nonzero number, or
- divide both sides by the same nonzero number.

These simple facts justify the familiar steps used to solve equations.

Example 2.1. Solve

$$5x - 7 = 2x + 8.$$

Solution. Subtract $2x$ from both sides and add 7 to both sides:

$$5x - 2x = 8 + 7.$$

Thus

$$3x = 15,$$

so

$$x = 5.$$

Can We Solve Every Linear Equation at Once?

Consider the general equation

$$ax + b = cx + d.$$

Subtract cx from both sides and subtract b from both sides:

$$(a - c)x = d - b.$$

If $a - c \neq 0$, we divide by $a - c$ and obtain

$$x = \frac{d - b}{a - c}.$$

This formula is useful, but the reasoning behind it is more important than memorizing it.

Can You Generalize? What happens when $a = c$? We shall answer this question later in the chapter.

2.2 Parentheses and Fractions

A linear equation may look complicated because of parentheses or fractions. The best strategy is usually to simplify first.

Example 2.2. Solve

$$x - (2x + 3) = 7.$$

Solution.

$$x - 2x - 3 = 7,$$

so

$$-x = 10,$$

and therefore

$$x = -10.$$

Example 2.3. Solve

$$\frac{x}{6} - 2(x - 1) = \frac{1}{4}.$$

Solution. The denominators are 6 and 4, whose least common multiple is 12. Multiply every term by 12:

$$2x - 24(x - 1) = 3.$$

Then

$$2x - 24x + 24 = 3,$$

so

$$-22x = -21.$$

Hence

$$x = \frac{21}{22}.$$

Professor Shen's Remark. Fractions often make an equation look harder than it really is. Before doing any algebra, ask whether multiplying by a common denominator will make the structure clearer.

Example 2.4. Solve

$$\frac{x}{11} - \frac{x}{12} = \frac{x}{2} + 1.$$

Solution. Multiply the equation by 132, the least common multiple of 11, 12, and 2:

$$12x - 11x = 66x + 132.$$

Thus

$$-65x = 132,$$

so

$$x = -\frac{132}{65}.$$

2.3 Equations That Only Look Nonlinear

Some equations contain fractions with x in a denominator, squares, or square roots. Such equations are not linear as written, but sometimes their structure allows them to reduce to a linear equation.

Whenever we multiply by an expression involving x , square both sides, or cancel a factor involving x , we must pay attention to values for which the operation may not be reversible.

Rational Equations

Example 2.5. Solve

$$\frac{2}{x-2} = \frac{3}{x-3}.$$

Solution. First note that

$$x \neq 2, \quad x \neq 3.$$

Cross-multiply:

$$2(x-3) = 3(x-2).$$

Therefore

$$2x - 6 = 3x - 6,$$

so

$$x = 0.$$

Since 0 is allowed in the original equation, it is the solution.

Example 2.6. Solve

$$\frac{x+4}{x-3} = 4.$$

Solution. Since $x \neq 3$, multiply both sides by $x-3$:

$$x+4 = 4(x-3).$$

Then

$$x+4 = 4x-12,$$

so

$$3x = 16.$$

Hence

$$x = \frac{16}{3}.$$

A Surprise from Squares

Example 2.7. Solve

$$r^2 = (5 - r)^2 + 7.$$

Solution. At first the equation looks quadratic. Expand the square:

$$r^2 = 25 - 10r + r^2 + 7.$$

The r^2 terms cancel:

$$0 = 32 - 10r.$$

Thus

$$r = \frac{16}{5}.$$

The important idea is not this particular answer. It is the cancellation of the squared terms.

Can You Generalize? Solve in terms of a, b , and c :

$$x^2 = (a - x)^2 + c.$$

For which values of a does this equation reduce to a linear equation with one solution?

2.4 Square Roots and Checking Solutions

An equation containing square roots requires extra care because the quantity inside a square root must be nonnegative.

Example 2.8. Solve

$$\sqrt{1 - x} = \sqrt{2x - 5}.$$

Solution. Before squaring, check the domains:

$$1 - x \geq 0 \quad \text{and} \quad 2x - 5 \geq 0.$$

Thus

$$x \leq 1 \quad \text{and} \quad x \geq \frac{5}{2}.$$

No real number satisfies both conditions. Therefore the equation has no real solution.

There is another lesson here. If we square both sides immediately, we obtain

$$1 - x = 2x - 5,$$

which gives $x = 2$. But $x = 2$ is not allowed in the original equation because both radicands are negative. Squaring can produce an **extraneous solution**.

Example 2.9. Solve

$$\sqrt{16 + (x + 1)^2} = \sqrt{25 + (7 - x)^2}.$$

Solution. Both sides are nonnegative for every real x , so we may square both sides:

$$16 + (x + 1)^2 = 25 + (7 - x)^2.$$

Expand:

$$16 + x^2 + 2x + 1 = 25 + 49 - 14x + x^2.$$

The x^2 terms cancel:

$$17 + 2x = 74 - 14x.$$

Thus

$$16x = 57,$$

so

$$x = \frac{57}{16}.$$

This is another example of a seemingly nonlinear equation becoming linear.

2.5 One Solution, No Solution, or Infinitely Many Solutions

Return to the general linear equation

$$ax + b = cx + d.$$

We obtained

$$(a - c)x = d - b.$$

Now three cases are possible.

Case 1: $a \neq c$

Then $a - c \neq 0$, and there is exactly one solution:

$$x = \frac{d - b}{a - c}.$$

Case 2: $a = c$ but $b \neq d$

Then the equation becomes

$$b = d,$$

which is false. Therefore there is no solution.

Case 3: $a = c$ and $b = d$

Then both sides of the original equation are identical. Every real number is a solution, so the equation has infinitely many solutions.

This classification is a good example of how mathematicians move from solving particular equations to understanding an entire family of equations.

Example 2.10. Find m so that

$$2x - 3(mx + 1) = 5 - 5x$$

has no solution.

Solution. Simplify:

$$2x - 3mx - 3 = 5 - 5x,$$

so

$$(7 - 3m)x = 8.$$

For the equation to have no solution, the coefficient of x must be 0 while the right side remains nonzero. Thus

$$7 - 3m = 0,$$

and therefore

$$m = \frac{7}{3}.$$

Example 2.11. Find m so that

$$3x + m = 3x + 7$$

has infinitely many solutions.

Solution. The x terms already agree. For both sides to be identical, we need

$$m = 7.$$

2.6 Parameters and Reverse Thinking

Sometimes a problem tells us a solution and asks us to find a parameter. Instead of solving for x , we substitute the known value of x .

Example 2.12. Suppose $x = 4$ is a solution of

$$\frac{1}{2}(x - 3) = \frac{1}{3}(x - m).$$

Find m .

Solution. Substitute $x = 4$:

$$\frac{1}{2}(1) = \frac{1}{3}(4 - m).$$

Multiply by 6:

$$3 = 2(4 - m).$$

Thus

$$3 = 8 - 2m,$$

so

$$m = \frac{5}{2}.$$

Professor Shen's Remark. Many competition problems become easier when you stop trying to solve the equation in the usual direction. Ask what information is already known and use it first.

What Have We Discovered?

In this chapter we learned more than a list of procedures for moving terms from one side of an equation to the other.

1. An equation is a balance, and valid operations preserve that balance.
2. The general equation $ax + b = cx + d$ leads to the formula

$$x = \frac{d - b}{a - c}$$

when $a \neq c$.

3. The same general equation may have one solution, no solution, or infinitely many solutions.
4. Parentheses and fractions often become easier after simplifying or clearing denominators.
5. Some equations that look nonlinear reduce to linear equations because higher-degree terms cancel.
6. Multiplying by an expression involving x or squaring both sides may require checking the final answer in the original equation.
7. A good problem solver does not only ask “How do I solve this equation?” but also “What happens for the entire family of equations?”

Solve the problem, then ask what the solution teaches you.

Exercises

The first ten problems are Core Exercises. They contain the essential work for this chapter and are appropriate for all students. Problems 11–20 are Challenge Problems. They are optional and are intended especially for students who enjoy difficult problems or wish to prepare for MATHCOUNTS and the AMC 8. All problems in this section have been newly written for this book.

Part A. Core Exercises

1. Solve

$$4x - 9 = 2x + 7.$$

2. Solve

$$x - [6 - (x + 1)] = -5.$$

3. Solve

$$\frac{3x}{4} + \frac{5x}{8} = 22.$$

4. Solve

$$\frac{x}{3} + \frac{x}{5} = \frac{16}{15}.$$

5. Solve

$$5 - 2(x - 3) = 3x + 1.$$

6. Solve

$$\frac{2x + 7}{x + 4} = 3.$$

7. Solve

$$\frac{3}{x - 1} = \frac{5}{x + 1}.$$

8. Solve

$$(x - 2)^2 = (x - 5)^2 + 9.$$

9. Find m if $x = 3$ is a solution of

$$2(x - 1) = x + m.$$

10. Find m so that

$$4x - 2m = 4x + 6$$

has infinitely many solutions.

Part B. Challenge Problems

11. Solve

$$\frac{1 - \frac{2}{x}}{1 + \frac{2}{x}} = 3.$$

12. Solve

$$\frac{x-2}{x-5} = \frac{x+4}{x+1}.$$

13. Find
- m
- so that

$$(m-2)x + 5 = 3x - 1$$

has no solution.

14. Find
- m
- so that

$$(2m+1)x - 4 = (m+5)x + 7$$

has exactly one solution.

15. Without solving for
- x
- and
- y
- separately, suppose

$$3x - 2y = 7$$

and

$$5x + 2y = 13.$$

Find x .

16. Solve

$$\sqrt{9 + (x+2)^2} = \sqrt{49 + (8-x)^2}.$$

17. The equation

$$ax + 4 = 6x + b$$

has the solution $x = 2$ for every value of a . Express b in terms of a .

18. A number
- x
- has the property that increasing it by 20% gives the same result as adding 18 to it. Find
- x
- .

19. Consider the following two equations:

$$(k-1)x + 2k = 3x + 8,$$

$$(k-1)x + 2k = 3x + 9.$$

For each equation, find the value of k for which it does not have exactly one solution. Decide whether the exceptional case gives no solution or infinitely many solutions.

20. Consider

$$ax + b = cx + d.$$

Write a short summary, in your own words, explaining exactly when this equation has one solution, no solution, or infinitely many solutions. Then create one example of each type.

Beyond the Chapter

These questions are optional. Their purpose is not speed, but discovery.

1. Starting from

$$ax + b = cx + d,$$

derive the solution formula yourself without looking back at the chapter. Then explain why the formula fails when $a = c$.

2. Consider

$$x^2 = (a - x)^2 + c.$$

Derive a formula for x in terms of a and c . What special case occurs when $a = 0$?

3. Create a rational equation that looks nonlinear but simplifies to a linear equation. Solve it and explain why any excluded values must be checked.
4. Write three different equations that all have $x = 5$ as their unique solution. Try to make the equations look as different from one another as possible.

Answer Key

1. $x = 8$.
2. $x = 0$.
3. $x = 16$.
4. $x = 2$.
5. $x = 2$.
6. $x = -5$.
7. $x = 4$.
8. $x = 5$.
9. $m = 1$.
10. $m = -3$.
11. $x = -4$.
12. No solution.
13. $m = 5$.
14. Any $m \neq 4$ gives exactly one solution.
15. $x = \frac{5}{2}$.
16. $x = 5$.
17. $b = 2a - 8$.
18. $x = 90$.
19. In both equations the exceptional value is $k = 4$. The first equation has infinitely many solutions; the second has no solution.
20. Answers will vary. One solution if $a \neq c$; no solution if $a = c$ and $b \neq d$; infinitely many solutions if $a = c$ and $b = d$.